LABORATORY PROJECT REPORT

DIGITAL LOGIC SYSTEM

EXPERIMENT 2

Section 1

Group 2

Semester 2 2025/2026

NO	NAME	MATRIC NO
1	AHMAD SYAHMI WAJDI BIN AHMAD SHUKRI	2310663
2	ALYA SYAFINA BINTI ABDUL RAZAK	2218084

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1.0 Introduction

This experiment focuses on interfacing a common cathode 7-segment display with an Arduino Uno to display numbers from 0 to 9 using digital logic. Pushbuttons are used to manually control the count, allowing for incrementing and resetting the display. The experiment covers key concepts in digital logic systems, such as basic logic gates, electronic circuit interfacing, and IC-based applications. Students will learn how to configure Arduino digital output pins, use resistors for current limiting, and write code to control the display. Additionally, the lab introduces a comparison between 7-segment displays, 12C LCDs, and LED matrix displays, highlighting their differences in coding and applications. This hands-on experiment helps students understand microcontroller programming, circuit design, and display interfacing in embedded systems. This experiment is anticipated to demonstrate basic digital logic control, binary-to-decimal conversion using code, and effective use of Arduino I/O for hardware interfacing.

2.0 Materials and Equipment

2.1 Electronic Components

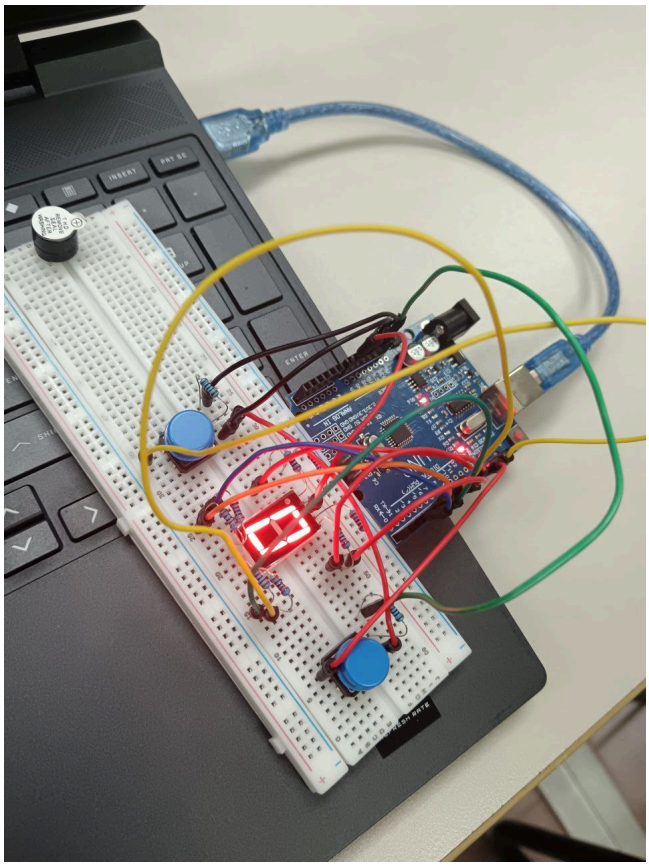
- Arduino Uno
- Common Cathode 7-Segment Display
- 220-Ohm Resistors (7x)
- Jumper Wires
- Breadboard
- Pushbutton

2.2 Equipment and Tools

- Arduino IDE
- USB Cable
- Power Supply (5V via USB or external source)

3.0 Experimental Setup

1. Connect each of the seven segments (a, b, c, d, e, f, g) of the 7-segment display to separate digital pins on the Arduino.
2. Attach the common cathode pin of the display to the ground (GND) pin on the Arduino.
3. For the pushbutton, connect one of its legs to a digital pin on the Arduino, another to the ground (GND) pin, and a third to the 5V pin through a resistor.



4.0 Methodology

1. Set up the Arduino Uno
2. Set up the code in Arduino IDE
3. Test the code
4. Code Snippet

4.1 Detail steps

1. Assemble the circuit based on the given setup instructions.

2. Upload the supplied Arduino program to your Arduino Uno board.

3. Launch the Serial Monitor in the Arduino IDE.

4. Press the increment button to raise the count. The 7-segment display should cycle through numbers 0 to 9 in order.

5. Press the reset button to bring the count back to 0

4.2 Programming codes

```
// Define the pins for each segment
const int segmentA = 3;
const int segmentB = 2;
const int segmentC = 8;
const int segmentD = 6;
const int segmentE = 7;
const int segmentF = 4;
const int segmentG = 5;
```

```
// Define button pins
const int buttonInc = 11;
const int buttonReset = 12;

// Variables
int number = 0;
bool lastIncState = HIGH;
bool lastResetState = HIGH;

// 7-segment display digits (for common cathode)
// HIGH = ON, LOW = OFF
const int digits[10][7] = {
  {HIGH, HIGH, HIGH, HIGH, HIGH, HIGH, LOW},   // 0
  {LOW, HIGH, HIGH, LOW, LOW, LOW, LOW},       // 1
  {HIGH, HIGH, LOW, HIGH, HIGH, LOW, HIGH},    // 2
  {HIGH, HIGH, HIGH, HIGH, LOW, LOW, HIGH},    // 3
  {LOW, HIGH, HIGH, LOW, LOW, HIGH, HIGH},     // 4
  {HIGH, LOW, HIGH, HIGH, LOW, HIGH, HIGH},    // 5
  {HIGH, LOW, HIGH, HIGH, HIGH, HIGH, HIGH},   // 6
  {HIGH, HIGH, HIGH, LOW, LOW, LOW, LOW},      // 7
  {HIGH, HIGH, HIGH, HIGH, HIGH, HIGH, HIGH},  // 8
  {HIGH, HIGH, HIGH, HIGH, LOW, HIGH, HIGH}    // 9
};
```

```
// Segment pins array
int segments[] = {segmentA, segmentB, segmentC, segmentD, segmentE, segmentF, segmentG};

void setup() {
  // Initialize 7-segment pins as OUTPUT
  for (int i = 0; i < 7; i++) {
    pinMode(segments[i], OUTPUT);
  }

  // Initialize buttons with internal pull-up
  pinMode(buttonInc, INPUT_PULLUP);
  pinMode(buttonReset, INPUT_PULLUP);

  // Start showing 0
  displayDigit(number);
}

void loop() {
  bool incState = digitalRead(buttonInc);
  bool resetState = digitalRead(buttonReset);

  // Increment button pressed (active LOW)
```

```
  if (incState == LOW && lastIncState == HIGH) {
    number++;
    if (number > 9) number = 0;
    displayDigit(number);
    delay(250); // debounce
  }

  // Reset button pressed (active LOW)
  if (resetState == LOW && lastResetState == HIGH) {
    number = 0;
    displayDigit(number);
    delay(250); // debounce
  }

  lastIncState = incState;
  lastResetState = resetState;
}
// Function to display a digit (0–9)
void displayDigit(int num) {
  for (int i = 0; i < 7; i++) {
    digitalWrite(segments[i], digits[num][i]);
  }
}
```

5.0 Data Collection

Button Press	Expected Display Output	Actual Display Output
1st	1	1
2nd	2	2
3rd	3	3
4th	4	4
5th	5	5
6th	6	6
7th	7	7
8th	8	8
9th	9	9
10th	0	0

Button Press	Expected Display Output	Actual Display Output
1st	1	1
Reset button	0	0

6.0 Data Analysis

The data obtained from the 7-segment display's response to the Arduino input plays an important role in assessing the effectiveness of the interface and control system. When the display shows the correct numbers for each button press, it indicates that the Arduino's digital signals are properly controlling each segment. This demonstrates a correct understanding and application of digital logic concepts as well as pin-level control using the microcontroller.

In summary, the data clearly shows how well the system achieved the experiment's objectives of using Arduino to control a 7-segment display through user input, providing practical proof of understanding basic digital electronics and microcontroller programming.

7.0 Results

The experiment effectively demonstrated how to interface a common cathode 7-segment display with an Arduino Uno. With the correct circuit connections and code implementation, the display was able to show numbers from 0 to 9 in sequence when the increment button was pressed, while the reset button returned the display to zero.

Each digit appeared clearly, and the transitions between numbers were smooth and immediate, showing that the control logic was working properly. The inclusion of 220-ohm resistors successfully limited the current, preventing issues such as overheating or flickering.

No noticeable delays or errors occurred during operation. The system responded quickly to button inputs, confirming that both the hardware setup and Arduino programming were functioning correctly.

8.0 Discussion

The results from our experiment confirm the correct operation of the digital logic system. The Arduino successfully controlled each 7 segment of the display through digital output pins, demonstrating the practical use of binary logic in visual output devices.

Each digit was formed by activating the appropriate combination of segments according to standard logic patterns. The system's accuracy and responsiveness prove that the circuit design and code structure were correctly implemented.

A possible minor issue could arise from loose breadboard connections or debouncing of pushbuttons, which might cause unintended multiple counts. This can be improved by adding a software debounce delay or using hardware filters.This experiment reinforces understanding of how logic gates, microcontroller I/O, and current-limiting resistors work together in digital display applications.

9.0 Conclusion

In conclusion, our experiment on the 7-segment display successfully demonstrated the implementation of digital logic control using an Arduino Uno. The display accurately showed numbers from 0 to 9 based on button inputs, confirming that the circuit wiring, resistor

configuration, and programming logic functioned as intended. This confirms the hypothesis that a properly interfaced microcontroller can produce correct numerical outputs through digital logic operations. The experiment provided a practical understanding of how binary logic is applied to real hardware systems. It also emphasized the importance of correct circuit connections, suitable resistor values, and reliable code structure in ensuring stable operation. Overall, the experiment achieved its objectives by reinforcing fundamental concepts in digital electronics and providing a solid foundation for future projects involving more complex display systems or logic-based microcontroller applications.

10.0 Recommendations

For future improvements of this experiment, it is important to carefully check all circuit connections to ensure that each component is properly connected to the correct Arduino pins. A clear understanding of the pin configuration is crucial for accurate coding and overall system performance. The code can also be improved by making it more concise and efficient, which helps save time and minimizes potential errors. It is recommended to build stronger programming skills through regular practice instead of depending too much on AI-generated solutions, as this leads to better understanding and flexibility when troubleshooting. Moreover, the troubleshooting methods learned in this experiment should be applied in future projects to quickly detect and fix wiring or logic issues, resulting in smoother progress and more reliable outcomes.

11.0 References

- Week 2: Digital Logic System: Basic Logic Gates, Electronic Circuit Interfacing, Basic ALU, 7 segment display, ICs based interfacing application (ver 2) https://drive.google.com/drive/folders/1QecGGN96D76UL2gwBjevFwiEXv2UEniO?usp=sharing

Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

Name: AHMAD SYAHMI WAJDI BIN AHMAD SHUKRI	Read	[/]
Matric Number: 2310663	Understand	[/]
Signature: SYAHMI	Agree	[/]

Name: ALYA SYAFINA BINTI ABDUL RAZAK	Read	[/]
Matric Number: 2218084	Understand	[/]
Signature: SYAF	Agree	[/]